A Granular Look into Firms' Cash Portfolios

by

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Abstract

This paper uses confidential Census data to provide a granular look into the U.S. firms' cash holding portfolios encompassing nearly four decades. The data provide information on short-term investment securities held in the portfolios, such as time deposits, commercial paper and government securities in addition to cash. The security-level information reveals that portfolios of the same size can have very different levels of liquidity and riskiness as the composition of securities varies considerably across firms and over time. Firms with strong precautionary motives tend to allocate more toward relatively more liquid and less risky securities. Firms actively rebalance their portfolios in response to changing economic conditions or idiosyncratic shocks to securities they hold. Event studies using shocks to Treasury securities and commercial paper shows firms shifting away from affected securities and simultaneously adjusting weights of other securities.

Keyword: Cash management, liquid assets, short-term investment securities

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1. Introduction

Firms' cash holdings have been studied extensively in the literature. The so-called cash holdings include various liquid assets in addition to cash (we call cash holdings "liquid asset holdings" throughout the paper to avoid confusion with the cash component of the holdings). Yet, the literature is very thin when it comes to explaining what type of securities firms hold in their liquid asset portfolios and how they rebalance their portfolios. This is mainly due to data limitations. Previous studies generally relied on a Compustat variable, "cash and short-term investments (CHE)," which is only split into two components, cash and cash equivalent (CH) and short-term investments (STIs).[1] Although STIs are defined as securities that are readily transferable to cash, these securities vary considerably in their liquidity as they have maturities up to one year and it can be costly to liquidate before maturity, especially under adverse market conditions. They are also riskier than cash and their riskiness varies over time due to idiosyncratic reasons or changing macroeconomic conditions. Unfortunately, Compustat does not have information on the identity and composition of the securities included in firms' STI holdings. Depending on the nature of securities in the STI bucket, a firm's liquid asset holdings can be quite risky and illiquid, contrary to the traditional view that the precautionary savings motive is central to corporate cash policy.

A few recent studies shed some light on this question as a new accounting guideline adopted in 2009 led firms to report details on their STI holdings in 10-Ks.[2] Most of these studies, which are based on very large firms or firms in a certain industry, highlight non-precautionary motives, such as corporate governance and tax considerations, as guiding firms to hold risky financial assets. However, one study using a broader sample finds that precautionary motives guide firms' risky liquid asset holdings, attributing the different finding to the sample coverage. Thus, it is unclear what would explain firms' decision to hold risky liquid assets in large panel data. In addition, these studies mostly focus on total risky liquid asset holding rather than the composition, so a question remains as to how firms determine the composition of STI holdings given that these securities vary in their riskiness and liquidity.

[1] Quarterly Compustat data provide the break-down into CH and STIs only starting from 2007.

[2] A detailed literature review is provided in the later part of the introduction.

In this paper, we provide a granular look into U.S. firms' STI portfolios using novel data: Confidential firm-level data underlying the U.S. Census Bureau's Quarterly Financial Report (QFR), a survey that collects income statements and balance sheets of U.S. firms. The QFR data report major STI securities at the firm level going back to 1977 on a quarterly frequency, including time deposits, commercial and finance company paper, U.S. Treasury and federal agency securities, state and local government securities, and other short-term financial investments securities. The high frequency nature of the data (quarterly) combined with a long time series coverage is especially useful for examining firms' rebalancing activities in response to changing economic conditions. To the best of our knowledge, this firm-level breakdown of STI holdings has not been utilized by researcher before. The aggregate version of the information has been used as an input in the Financial Accounts of the United States (formerly known as the Flow of Funds), published by the Federal Reserve Board.

We match the QFR micro data to Compustat data based on the company name and other firm characteristics to obtain additional firm characteristics information from Compustat. We exclude firms that never held STIs in the sample period, as our focus is to understand firms' STI portfolios. The resulting data set contains 141,000 observations encompassing 4,900 unique U.S. firms at the quarterly frequency between 1977:Q1 and 2014:Q1.[3] Our final sample is comparable to Compustat: Figure 1a shows that total liquid assets relative to total assets constructed using Compustat (blue line) and the same variable constructed using the QFR (red line) are remarkably similar in terms of their levels and changes over time. However, when it comes to the composition of liquid assets, the two data sets depict different patterns. Figure 1b shows that the ratio of STIs to total liquid assets constructed using Compustat plunges abruptly in 1988 due to a change in accounting guidelines while the variable constructed using the QFR is not susceptible to such changes in accounting treatments and provides a consistent treatment over time.[4]

We use the matched sample to answer three important questions. First, we examine the composition of STI holdings for a broad set of firms encompassing nearly four decades. Firms

[3]The number of observations and the number of firms are rounded per the Census disclosure guideline.

[4]The spike in the Compustat measure is an artifact of the adoption of Statement of Financial Accounting Standards guideline number 95, which gives firms more leeway in classifying liquid assets as either cash or marketable securities (Gao, Whited, and Zhang (2021)).

in our sample hold about 32% of their liquid assets in the form of STIs on average, but there is considerable time-series variation with the share of STIs declining gradually from over 40% in the early 1980s to under 30% in recent decades with some bumps along the way. Time deposits led the decline, decreasing from 17% of total liquid assets in the early 1980s to around 7% in the 2000s. There were also considerable cross-sectional variations in the composition of STI portfolios. For instance, STIs relative to total liquid assets ranged from 24% for the wholesale/retail/services sector to 42% for the health sector. The granular, security-level data reveal that industries with similar STI/CHE ratios (that is, STIs scaled by total liquid assets) do not necessarily have similar STI compositions, implying that the risk and liquidity profiles of their STI portfolios can be different from what is inferred from the STI/CHE ratio. The composition of STI portfolios also differs considerably across firm characteristics.

Second, we utilize variations at the individual security level to gain insights into what motives drive firms to hold a different mix of STI securities and how the decision interacts with firm characteristics and market conditions. The traditional literature documents that precautionary motives are an important driver in firms' decision to hold liquid assets. However, the driver of the allocation decision within STI portfolios can be different, and recent studies point out some non-precautionary motives to hold risky liquid assets in the case of very large firms. We examine whether the allocation decision is also guided by precautionary motives in a large panel data set. Because of the differences in liquidity and riskiness across STI securities, their sensitivities to firm-level determinants of the STI portfolio composition can differ considerably from what was previously documented in the literature using Compustat's total liquid asset variable (CHE). We regress total STIs as well as individual STI components (all scaled by total liquid asset holdings) on firm-level determinants of liquid asset holdings that are generally considered in the literature. Firms with strong precautionary motives hold more cash and less STIs as a fraction of total liquid assets. Within STI portfolios, these firms also tend to have more securities that are relatively more liquid and less risky, consistent with precautionary motives. For example, these firms hold less time deposits such as negotiable certificates of deposit (NCDs), which can be more illiquid due to their relatively longer maturities and illiquid secondary markets. These variations within STI portfolios across firms

mean that two firms with liquid asset holdings of the same size can be quite different in terms of their liquidity and riskiness.

Third, we examine how firms rebalance their STI holdings over time as liquidity and riskiness of STI securities fluctuate. Specifically, we consider two shocks that directly affected individual securities to examine the effect on both the affected securities and the overall STI composition. One event we consider is a financial commercial paper (CP) run that occurred in 2008:Q3. It was triggered by Lehman's default on its CP, but quickly spread to other CP, resulting in a major disruption in the CP market. Comparing the period surrounding the CP run, we find that firms cut their CP holdings considerably. They also sold state and local government securities and some other securities while buying more time deposits, likely in an attempt to reduce risk. These rebalancing activities change the composition of STI portfolios, which is not observed from the STI/CHE ratio. The other event we consider is the U.S. debt-ceiling crisis in 2013:Q4. T-bill rates rose sharply as the debt ceiling deadline approached due to uncertainty over when T-bill investors would be paid. We find that firms cut their Treasury holdings as expected, but this reduction was met with a comparable increase in CP holdings. As a result, the net effect on total STI holdings was negligible. These event studies clearly demonstrate that firms responded to the shocks and rebalanced their STI portfolios, consistent with precautionary motives.

Even in the absence of changes in liquidity and riskiness of STI securities, firms may rebalance their STI portfolios as their desired level of liquidity and riskiness change over time. For instance, liquidity needs based on precautionary motives may be lower in recessions as sales slow down and new investment opportunities decline. We consider two measures to capture changing economic conditions. First, we consider a recession dummy variable. We find that firms do not change the size of STI holdings relative to total liquid assets significantly during recessions, controlling for firm characteristics. However, a granular look into the security-level analysis reveals that firms increase time deposits and reduce Treasuries, state/local government securities, and other STI securities. Second, we consider an interest rate variable. Increases in interest rates typically coincide with strong economic growth. We find that firms increase STI holdings relative to total liquid assets when interest rates are higher, as precautionary motives are likely lower when economic conditions are generally more favorable. The

security-level examination shows that the increase is broad-based across almost all STI securities including deposits, CP, Treasuries, and other securities. Overall, firms appear to rebalance their portfolios in a manner consistent with precautionary motives.

Another important takeaway is that changes in the STI composition can result in very different implications for riskiness and liquidity of liquid asset portfolio than what can be inferred from just observing the aggregate STI holdings. Riskiness and liquidity can vary considerably across different STI securities. For example, NCDs, a popular time deposit instrument, and CP trade at a spread above Treasuries of the same maturity because investors view these instruments as carrying some credit risk, which Treasuries do not. Secondary markets for these instruments are also generally more illiquid, making these securities more costly to convert to cash to meet unexpected cash needs, especially when remaining maturity is long. In the CP market, secondary-market trading accounts for less than 10% of total CP trading (Gross, Li, and Wang (2022)). NCDs can be viewed as a more illiquid investment choice than CP because they offer longer maturities on average.[5] For example, according to Depository Trust & Clearing Corporation data, only about 20% of NCDs outstanding had maturities of one month or less while about half of CP outstanding was maturing within a month in August 2024.

Riskiness and liquidity also fluctuate over time. For instance, the CP market seized up during the Global Financial Crisis and the COVID-19 crisis and only recovered following interventions from the Federal Reserve. Auction-rate securities (ARS) failures in 2008 are another example. ARS are treated as a liquid, short-term instrument because ARS interest rates are set periodically in auctions, although the underlying securities are essentially long-term debt. But the auction market had widespread failures in early 2008, leaving many investors with sudden illiquid holdings.[6] Treasury securities are not immune from shocks, as discussed above in the event study regressions.

Our work adds to the cash holdings literature, in particular the recent effort to identify the composition of firms' liquid asset holdings and to examine its determinants and implications. Earlier work by Brown (2014), Azar, Kagy, and Schmalz (2016), and Cardella, Fairhurst, and

[5] When compared on the same maturity, CP is generally considered slightly less liquid than CDs (Fabozzi and Man (2012)).

[6] "Rain pulls out on new corner of debt market," Wall Street Journal, February 14, 2008.

Klasa (2021) relied on Compustat data to compare the determinants of cash and those of marketable securities. However, more granular liquid asset information became available starting in 2009 following the adoption of Statement of Financial Accounting Standards guideline #157, which requires firms to report the composition and fair value of their investment securities. Several recent studies utilized this information. Duchin et al. (2017) extract this information from 10-Ks for nonfinancial S&P 500 firms between 2009 and 2012 and document that the drivers of risky financial assets are different from those of total liquid assets: While the overall size of liquid asset portfolios increases with demand for precautionary savings consistent with Opler et al. (1999) and Bates, Kahle, and Stulz (2009), the fraction of risky financial assets is higher for financially unconstrained firms and poorly governed firms. Darmouni and Mota (forthcoming) also examine very large firms. Analyzing bond holdings as well as overall holdings for the 200 largest firms between 2000 and 2021, they document that tax optimization and reaching for yield are the main reasons behind the rise in these firms' corporate bond holdings up to the 2017 tax reform and the subsequent decline. Chen and Duchin (2024) complement these studies by considering a different subset of firms. They obtain the information for oil and gas companies between 2011 and 2016 to examine the behavior of distressed firms around the 2014 oil price crisis. They document that distressed firms increase risky financial asset investment substantially, likely using financial investments as a conduit for risk-taking. Huang and Sacchetto (2023) question whether precautionary motives can explain the risky financial asset holdings in a broader cross section of firms. They extract information from 10-Ks for about 2,800 firms between 2009 and 2018. They document that firms tend to hold more risky financial assets when their financing deficit—defined as investment needs in excess of cash flows—positively correlates with returns on financial assets. They interpret that this is consistent with precautionary motives, as financial asset holdings improve these firms' alignment over time between internal cash flows and investment opportunities.

Our work complements these studies well. First, we examine a large panel data set (the majority of Compustat firms over four decades), which helps us find general patterns. Second, our analysis is done at the security level, which highlights that liquidity and riskiness can vary considerably for two STI portfolios of the same size. The studies using information from 10-Ks usually allocate identified securities into two buckets, safe and risky assets and mostly

focus on the overall risky assets rather than the compositions. Third, we exploit the relatively higher-frequency nature of the data (quarterly) to track firms' rebalancing activity closely, while these studies usually use annual frequency information.[7] The higher frequency allows us to better observe firms' response around an event, such as a debt ceiling crisis or a CP run, which is harder to detect in annual frequency data.

2. Data and Summary Statistics

2.1. Quarterly Financial Report (QFR)

We obtain firms' detailed liquid asset holdings information from the QFR data. The QFR is a survey of firms conducted each quarter by the U.S. Census Bureau. Data collected by the QFR are used as an input in various official statistical publications, including the national income and product accounts. The aggregated version of the data has also been broadly used in research, mainly in the macroeconomics literature.[8] The only paper utilizing the micro-level QFR, to the best of our knowledge, is Crouzet and Mehrotra (2020), but they do not use the liquid asset information, as their focus is on a macroeconomic question of the relationship between firm size, cyclicality, and financial frictions. However, the liquid asset information aggregated at the national level is used as an input in the Financial Accounts of the United States, published quarterly by the Federal Reserve Board. Figure 2 highlights the variables in the nonfinancial corporate business table of the Financial Accounts that are constructed using the liquid asset information from the QFR. We use the micro-level information to break down firms' STI portfolios in five categories as follows:

- Time deposits including NCDs, savings accounts, and other interest-bearing cash deposits ("DEP")
- Commercial and finance company paper ("CP")

[7] An exception is one subsample analysis in Darmouni and Mota (forthcoming), for which they collect quarterly data for a short time period.

[8] For example, see Gertler and Gilchrist (1994), Nilsen (2002), and Oliner and Rudebusch (1996).

- U.S. Treasury and federal agency securities subject to agreements to sell or due in one year or less, which we call Treasuries for brevity ("TRS")
- State and local government securities due in one year or less ("SLGOV")
- All other short-term financial investment securities ("Other")

The "other" category is a residual category capturing all STI securities that are not captured in the previous four categories. This category contains some risky securities such as stocks, commodities, and options. It also includes foreign securities. The QFR reports a separate category called "foreign securities due in one year or less," but it constitutes a negligible fraction (about 0.01% of liquid assets), so we include it in the "other" category. A detailed description of STI variables, provided by the QFR, is in appendix A.

2.2. Sample Construction

While the QFR provides rich details on the composition of STI securities, it lacks information on many firm characteristics variables needed for the regression analysis. Also, the definition of some variables changed over time, making it difficult to track them over time. Thus, we obtain other firm-level variables from Compustsat. To do so, we match the QFR to Compustat. The two data sets are matched mainly on the company name. Because the QFR microdata are not linked across time, we match the two data sets each quarter.[9] We further match and validate using employer identification number, the headquarter location, assets, and sales. The QFR data are reported on calendar quarters, while the Compustat data are reported on fiscal quarters, so the match is done on closest quarter-ends. We exclude firms that never held STIs in the sample period, as our focus is to understand firms' STI portfolios. The sample size is only slightly larger if we include these firms. We also exclude banks and utilities following the convention. The resulting data set contains about 141,000 observations encompassing 4,900 unique U.S. firms at a quarterly frequency between 1977:Q1 and 2014:Q1.

[9]The QFR data contain a survey identifier, but its encoding format changed over time, making it difficult to follow a firm over time.

2.3. Summary Statistics

Panel A of Table 1 summarizes the composition of liquid asset holdings for firms in our sample. Figure 3 provides visual representation. The left chart shows cash ("CH") and STIs as a fraction of total liquid asset holdings. The right chart further breaks down STIs into five individual components. Firms hold about 32% of their liquid assets in the form of STIs, of which deposits make up about 10%, CP about 6%, Treasuries about 4%, state and local government securities about 1%, and other securities about 9%. Turning to the time-series aspect of STI portfolios, Figure 4 depicts how each STI component as a share of total liquid assets ("CHE") fared over time. STIs relative to CHE shows a declining trend over time¡ with time deposits taking the largest share of the decline. This is consistent with the general trend of certificates of deposit (CDs), a popular time deposit instrument. CDs were a top money market instrument in the 1980s, but they have declined as a fraction of the money market as more instruments have become available over time; however, there remains a relatively large market for CDs (Stigum and Crescenzi (2007)).

Panel B of Table 1 summarizes other firm characteristics variables, taken from Compustat. Note that the sample size is down to 99,500 in Panel B because we only keep the observations for which all variables are non-missing to help comply with the Census disclosure requirement. Of these, four variables (CAPX/assets, R&D/Sales, dividend dummy, and acquisition/assets) do not have quarterly observations until the late 1980s. Thus, for the early years when quarterly observations were unavailable, we construct them using the information from annual Compustat data. Section 3.5 describes the construction of these variables in detail and provides robustness checks.

3. Empirical Analysis

3.1. Cross-Sectional Variations in Liquid Asset Composition

This section examines variations in the liquid asset composition across industries and firm characteristics. First, Table 2 summarizes the composition for 9 industries according to the Fama-French 10 industry classification, excluding utilities (utilities and financial firms are excluded from the sample following the convention). There are considerable differences across industries in the size of STIs relative to total liquid asset holdings: The STI/CHE ratio ranges from 24% for the wholesale/retail/services sector to 42% for the health sector.

Interestingly, a granular look into the STI portfolios reveals that industries with similar STI/CHE ratios do not necessarily have similar risk and liquidity profiles. For instance, consider the two sectors that hold the least amount of STI relative to CHE: The wholesale/retail/services sector and the energy sector. While the two have similar STI/CHE ratios (24% and 27%, respectively), the latter (the energy sector) allocates almost twice as much to time deposits as the former, with time deposits accounting for 13% of total liquid assets for the latter while accounting for only 7% for the former. By contrast, the technology sector and the health sector, which are also similar in their STI/CHE ratios (39% and 42%, respectively), have roughly the same compositions across the five STI security categories. These two sectors not only hold the largest amount of STI relative to total liquid assets, but also hold the largest fractions of relatively risky securities within the STI bucket: They hold the most CP, state/local government securities, and other securities, which are generally riskier than time deposits and Treasuries.

Table 3 splits the sample into quartiles according to various firm characteristics. Precautionary motives are stronger for financially constrained firms, and firm size and age are particularly useful predictors of financial constraint levels (Hadlock and Pierce (2010)). First, we split the sample according to firm size. While the relationship between STI holdings and firm size is not monotonic, the largest firms—those in the fourth quartile (Q4)—tend to hold more STI securities, with a ratio of STI/CHE of 0.333. This is consistent withe the precautionary motives that financially less constrained firms hold more STIs as a fraction of the total liquid

assets. Another noticeable pattern is that the STI composition varies considerably across the size quartiles. For example, the difference in CP holdings between Q1 and Q4 is quite large; the largest CP/CHE ratio is 0.071 in Q4 and the smallest is 0.038 in Q1, resulting in the difference of 0.032. In fact, it is larger than the difference in total STI holdings (0.028), although CP accounts for less than 20% of total STI holdings. This provides evidence that firms actively manage their STI portfolios. Different STI compositions have different liquidity and risk implications, which cannot be inferred by observing the overall STI/CHE ratio.

The second split is based on firm age, defined as the number of years that a firm remained in the Compustat data. Unlike the split based on firm size, we see no clear pattern associated with firm age. For example, firms in the Q4 bucket (oldest firms) hold the least amount of STIs relative to total liquid assets, with a STI/CHE ratio of 0.305, while firms in Q1 and Q3 hold the most, with ratios of 0.320 and 0.326, respectively. In addition, the magnitude of variations is fairly small.

The third split is based on R&D/Sales. The first and second quartiles are combined to capture the firms reporting no R&D expenses, which constitute about half of the observations.[10] Very different patterns emerge across the R&D quartiles. Both total STIs as well as individual STI components are monotonically increasing from the least-R&D-intensive groups (Q1 and Q2) to the most-intensive group (Q4). In addition, the magnitude in variations across the quartiles is very large: The most-R&D-intensive firms (Q4) hold 39.8% of liquid assets in the form of STI securities while the least-R&D-intensive firms (Q1 and Q2) hold merely 27%. Previous studies document that more-R&D-intensive firms tend to hold more liquid assets (e.g., Falato et al. (2022)). We further show that these firms hold a bigger fraction of liquid assets in the form of STIs (as opposed to cash). High R&D firms (Q4) hold more STI securities across all five STI categories. In particular, compared to low-R&D firms, high-R&D firms hold a lot more "other" securities (the residual category), which contain risky securities such as stocks, commodities, and options.

Taken together, the examination of cross-sectional variations across industry and firm characteristics reveals that there are considerable variations in the composition of STIs across

[10]Note that R&D is set to zero when the observation is missing, which is a typical treatment in the literature.

firms, highlighting that firms actively manage their STI portfolios. In the next section, we investigate how various firm characteristics explain the different composition of STI portfolios across firms.

3.2. Sensitivity to Firm Characteristics

We examine whether well-known determinants of cash holdings (i.e., total liquid asset holdings) also explain the composition of liquid assets. To do so, we estimate the standard specification in the cash holdings literature as follows:

$$Y_{i,t} = \alpha + \beta' X_{i,t} + \gamma_j + \theta_q + \varepsilon_{i,t}, \quad (1)$$

where the dependent variable ($Y_{i,t}$) is total liquid asset holdings or its components. $X_{i,t}$ are the firm-level determinants of liquid asset holdings. γ_j captures industry fixed effects. The specification also includes quarter fixed effects (θ_q) to control for seasonality.

We first examine whether our sample is comparable to the Compustat universe by using liquid asset holdings scaled by total assets (CHE/assets) as the dependent variable. The first column of Table 4 reports the results. The coefficient estimates are remarkably similar to those in a similar regression by Bates, Kahle, and Stulz (2009) in terms of the sign and significance despite the differences in the sample coverage. That is, the previously documented sensitivity of liquid asset holdings to firm characteristics remains similar in our sample. Having checked that our sample is comparable to the Compustat universe, we next exploit the granularity of our data and examine the composition of liquid assets. Column (2) of Table 4 examines total STIs (removing the cash component from liquid asset holdings). Columns (3) to (7) zoom in on individual STI components.

The second column shows that, controlling for liquid asset holdings relative to total assets, the sensitivity of STIs scaled by total liquid asset holdings (STI/CHE) is very different from the pattern in column (1): Coefficient estimates have opposite signs for almost all firm characteristics variables (industry sigma, market to book, firm size, NWC/assets, CAPX/assets, R&D/sales, and dividend dummy). For example, industry sigma is positively correlated with

CHE/assets in column (1) but negatively correlated with STI/CHE in column (2). That is, firms with greater cash flow risk hold more liquid assets as a fraction of total assets, consistent with precautionary motives in the literature. Within a liquid asset holdings portfolio, precautionary motives appear to lead firms to hold more cash and less STI securities as a fraction of total liquid asset holdings, likely because STI securities are less liquid and more risky than cash. Zooming in on the STI portfolio, we find that firms with greater cash flow risk cut time deposits (DEP/CHE) the most among STI securities, with half of the reduction in STIs coming from deposits: The coefficient estimate of -0.391 in column (3) is about half of the coefficient estimate (-0.783) in column (2). The other STI securities in columns (4) to (7) contribute much less to the reduction, ranging between -0.189 and -0.014. The relative change in time deposits is still large in magnitude when we consider the summary statistics that time deposits constitute close to one-third of STI security holdings on average (Table 1). This disproportionally larger reduction is consistent with firms with strong precautionary motives rebalancing their STI portfolios by moving away from relatively less liquid securities. As discussed earlier, time deposits such as NCDs can be an especially illiquid investment choice due to their relatively longer maturities in addition to having illiquid secondary markets.

The literature has a similar prediction for the dividend dummy variable: Precautionary motives are weaker for firms that pay dividends, as they are likely to be less risky and have greater access to capital markets, leading them to hold less cash relative to total assets. Consistent with the prediction, the coefficient in column (1) is negative, meaning that dividend payers hold less liquid assets relative to their total assets. These firms then go on to allocate less to cash and more to STI securities within their liquid asset portfolios, as shown in column (2). Regressions on individual STI securities in columns (3) to (7) show that dividend payers increase CP/CHE the most among STI securities, with the coefficient estimate in column (4) being the largest. This contrasts with the average allocation pattern reported in Table 1, which shows that time deposits receive the highest allocation among STI securities, accounting for over 10% of liquid assets, while CP accounts for only about 6%. That is, dividend payers appear to allocate larger fractions of their liquid assets to the type of STI securities that are relatively more risky but deliver higher returns. This allocation decision is consistent with the pattern predicted by precautionary motives.

The analysis above is based on a specification with industry fixed effects. Thus, the documented differences in sensitivities across STI securities can come from differences across firms and within firms as well as across time. We next focus on within-firm differences by including firm fixed effects (γ_i) as below. We also replace quarter fixed effects with quarterly frequency time fixed effects (θ_t) to control for time-series variations:

$$Y_{i,t} = \alpha + \beta' X_{i,t} + \gamma_i + \theta_t + \varepsilon_{i,t}. \quad (2)$$

Table 5 report the estimation results. We see that some of the significant patterns in Table 4 disappear once these fixed effects are introduced. For example, coefficients on industry sigma are no longer significant, suggesting that the significant loading in Table 4 likely came from differences across firms or time. Still, we see a considerable variation in sensitivities to firm characteristics variables across individual STI components. For example, leverage had a negative relationship with STI variables across the board in Table 4, which continues to be the case in Table 5. That is, firms reduce liquid asset holdings relative to total assets and lower the allocation to STI securities within their liquid asset portfolios as their reliance on debt financing increases. The leverage-related reduction in STIs is most pronounced in time deposits, which accounts for half of the reduction: The coefficient estimate in column (1) is -0.168, about twice of that in column (2) of -0.076. The other STI securities in columns (3) to (6) contribute much less to the reduction, with the coefficient estimates of leverage ranging between -0.014 and -0.027. As discussed earlier, the disproportionally larger reduction in time deposits is consistent with firms with more liquidity needs allocating less to time deposits within their STI portfolios, as these deposits tend to be more costly to convert to cash to meet unexpected cash needs.

The analysis so far shows that firms are attentive not only to the size of total liquid asset holdings, but also to the allocation between cash and STIs and, furthermore, the allocation among different STI securities. The firm characteristics tied to total liquid asset holdings also appear to guide these allocation decisions, helping to explain the variations across firms with different precautionary needs as well as the variations within firms over time. The within-firm

variations also mean that firms actively manage the composition of their liquid asset portfolios. In the next section, we further examine the time-series aspect of the variations.

3.3. Event Regressions

The relative liquidity and riskiness of STI securities fluctuate over time. Also, even in the absence of relative changes in liquidity and riskiness of STI securities, firms may rebalance their STI portfolios as their desired level of liquidity and riskiness change for firm-specific reasons or due to changes in economic conditions. In this section, we consider two events that affected the riskiness and liquidity of some securities more than others. We examine the effect of the shocks on the directly affected securities as well as the extent that the disruption was transmitted to other STI instruments. We also examine changes in macroeconomic conditions in the next section (section 3.4).

The first event we consider is a financial CP run that occurred in 2008:Q3, which is separate from the asset-backed CP market collapse in 2007 at the onset of the financial crisis. Lehman's default on its CP in September 2008 led to a run on the money market funds that were holding Lehman CP. Lehman's default was viewed as a signal that CP was far riskier than previously thought. The run quickly spread to other money market funds that were holding CP, resulting in a major disruption in the CP market. This event is described in more detail in Kacperczyk and Schnabl (2010). To capture this event, we define four quarters starting in 2008:Q3 as the event window and compare it to the previous four-quarter period using our baseline specification. Given the short time window, we include quarter fixed effects and industry fixed effects rather than time fixed effects and firm fixed effects.

Panel A of Table 6 reports the results. As expected, column (3) shows a large negative coefficient for CP/CHE, indicating that firms reduced holdings of CP relative to total liquid assets by 0.014. This change is economically meaningful. An average firm, which holds 5.9% of liquid assets in the form of CP according to Table 1, is reducing its CP holdings to 4.5%. One may argue that this reduction is not entirely voluntary, as financial CP outstanding dropped amid the run, lowering the supply of CP available for purchase. However, the spreads

on CPs over the federal funds rate increased at the same time, indicating diminished investor appetite for CP even with the reduced supply. Thus, it is unlikely that firms reduced holdings of CP involuntarily. Firms went on to reduce holdings of other relatively risky STI securities as well: State and local government securities (column (5)) and other STI securities (column (6)) also declined significantly as a fraction of total STI holdings. At the same time, they increased time deposits (column (2)) by about the same amount as they decreased CP.[11] On net, the STI/CHE ratio declined by 0.029 in the period following the event relative to the pre-event period. That is, an average firm, which allocates about 32% of liquid assets to STIs, reduced its STI holdings by about 3 percentage points. This is the net effect of reducing CP and other relatively risky securities by about 4 percentage points (the sum of the changes in CP/CHE, SLGOV/CHE, and Other/CHE) and increasing time deposits by about 1 percentage point.[12] Thus, the granular look at the STI portfolio reveals that firms reduced relatively risky securities more than what one can infer from the change in STI/CHE by rebalancing their STI holdings extensively, encompassing almost all STI security types.

We next consider a shock to the T-bill market: The debt ceiling crisis in 2013:Q4. As Ozdagli and Peek (2013) describe in detail, market participants anticipated a crisis as they approached the October 17 deadline when the borrowing authority of the United States would be exhausted. Accordingly, T-bill rates rose sharply due to uncertainty over when T-bill investors would be paid. Even after the resolution of the crisis later in October, the market was still uneasy, as market participants were anticipating the next debt ceiling deadline in early 2014. The nature of this crisis is somewhat different from the first event above in that the concern was mainly about liquidity—about when investors would be paid—rather than about whether they would be paid in full. To capture the Treasury market shock, we define two quarters starting in 2013:Q4 as the event window and compare it to the previous four quarters. The setup is the same as in the first event regression, but the post-event window is comprised of only two quarters this time because the sample ends in 2014:Q1.

[11] The default risk of NCDs declined as FDIC increased the standard maximum deposit insurance amount per depositor/bank from $100,000 to $250,000 in October 2008. Also, the composition of NCD issuers was likely skewed a little bit more towards banks perceived as too-big-to-fail than CP issuers.

[12] The coefficients do not exactly sum up to match due to rounding errors.

Panel B reports the result. As expected, firms reduced the fraction of Treasury securities relative to total liquid assets. The coefficient estimate (-0.007) in column (4) indicates that an average firm, which holds 4.2% of liquid assets in Treasuries, reduced its Treasury holdings by 0.7 percentage point. While the decline is statistically significant, its magnitude is not very large. This may be partly because the crisis did not affect all Treasuries but only those maturing around the debt ceiling deadlines. The reduction in Treasuries was met with a comparable increase in CP as shown by the coefficient estimate of 0.009 in column (3). This pattern is consistent with the asset substitution channel discussed in Ozdagli and Peek (2013), where investors would increase demand for CP to replace their reduced demand for Treasury bills. As a result, the net effect on total STI holdings was negligible. The coefficient estimate of 0.002 in column (1) is statistically insignificant and economically small. The result clearly demonstrates that firms responded to the shock and rebalanced their liquid asset compositions to improve liquidity of their STI portfolios. However, one cannot detect the change in perceived liquidity in the absence of granular information on the STI composition.

3.4. Macroeconomic Conditions

Next, we consider changes in broad economic conditions, which can induce firms to rebalance their STI portfolios as their desired level of liquidity and riskiness change over time. For instance, liquidity needs based on precautionary motives may be lower in recessions as sales slow down and new investment opportunities decline. We consider two measures to capture changing economic conditions: recession indicator and interest rate variables. First, a recession variable is set to one if a given quarter falls to a recessionary period according to the NBER business cycle classification and zero otherwise. Second, an interest rate variable is defined as the three-month Treasury security daily rate averaged each quarter. We add these measures to the baseline specification. The regressions use firm fixed effects and quarter fixed effects, as time fixed effects would subsume the macro variables. Table 7 report the results.

Panel A uses the recession indicator. Column (1) shows that the coefficient on the recession dummy is insignificant when the STI/CHE ratio is considered. That is, when STIs are examined in aggregate, firms do not seem to rebalance their portfolios in recessions, control-

ling for firm characteristics. However, a granular look into the security-level analysis tells a different story: Firms do rebalance their STI portfolio actively. They increase time deposits as shown in column (2) and reduce Treasuries, state/local government securities, and "other" STI securities as shown in columns (4) to (6). During recessions, firms might be more inclined to hold time deposits such as NCDs, although these securities generally have longer maturities and, hence, are relatively more illiquid, because firms' needs to fund investment, new or existing, tend to decline. On the other hand, they might be less inclined to hold government securities or "other" securities, which might be more liquid but also more risky, as their prices can be more volatile during recessions. Such active rebalancing has important implications for liquidity and riskiness of STI portfolios, but it is hard to detect if one can only observe the total liquid asset holdings and not the composition of the holdings.

Panel B uses the interest rate variable. The result shows that firms actively rebalance in response to the changing interest rate environment. Column (1) shows that firms increase STIs and decrease cash as a fraction of their total liquid assets when interest rates are higher, which allows firms to take advantage of higher rates, generating higher interest income. Further looking into the individual STI securities in columns (2) to (6), we see that the increase is broad based, with coefficients being all positive and mostly significant. In this case, rebalancing of individual securities was all done to the same direction. Still, one cannot infer the changes in the relative weights of individual STI securities if one can only observe the aggregate STI holdings and not the composition. For example, a relatively larger increase in time deposits has different implications for riskiness and liquidity of STI portfolios than a relatively larger increase in "other" securities.

3.5. Robustness Checks

Our analysis is based on quarterly frequency data, but some of the firm-level control variables are available only at an annual frequency until the late 1980s. These variables are CAPX/assets, R&D/Sales, dividend dummy, and acquisition/assets. Thus, in our main analysis, we use the information from annual Compustat for the early years when quarterly observations were unavailable. For the dividend dummy variable, we treat firms to be dividend

payers if annual Compustat data indicate that they pay dividends. For flow variables (CAPX, R&D, and acquisition), we allocate the annual flows equally across the four quarters within a year. However, it is possible that these values fluctuate from quarter to quarter. Thus, as a robustness check, we re-run regressions excluding these four variables in Table 8. We find that the results are similar to those in Table 5.

4. Conclusion

We use confidential Census data to provide a granular look into liquid asset portfolios for a broad sample of firms, encompassing nearly four decades. What the literature calls cash holdings includes various STI securities such as CP and government securities in addition to cash. We document that the composition of these securities varies considerably across firms and over time, likely in response to changing riskiness and liquidity needs. The firm characteristics tied to total liquid asset holdings also guide the allocation decision within STI portfolios, helping explain the variations in the composition across firms with different precautionary needs as well as the variations within firms over time. The within-firm variations suggests that firms actively manage the composition of their liquid asset portfolios. The high-frequency nature of the data (quarterly) combined with the long time series reveals that firms rebalance their STI holdings as liquidity and riskiness of STI securities fluctuate due to macroeconomic conditions or idiosyncratic shocks to some securities. Aggregate STI holdings do not reveal rebalancing within the portfolios, which has implications for risk and liquidity of liquid asset portfolios. That is, two firms with the same ratio of STI to total liquid assets can differ a lot in terms of liquidity and riskiness depending on the weight of securities in their STI portfolios. Overall, we find that firms with strong precautionary motives tend to allocate more toward relatively more liquid and less risky securities.

References

Azar, Jose, Jean-Francois Kagy, and Martin Schmalz. 2016. Can Changes in the Cost of Carry Explain the Dynamics of Corporate "Cash" Holdings? The Review of Financial Studies 29(8), 2194–2240.

Bates, Thomas, Kathleen Kahle, and René Stulz, 2009. Why Do U.S. Firms Hold So Much More Cash Than They Used To? Journal of Finance 64, 1985–2021.

Brown, Craig. 2014, Marketable Securities: Storage or Investment? Working paper. National University of Singapore.

Cardella, Laura, Douglas Fairhurst, and Sandy Klasa. 2021. What Determines the Composition of a Firm's Cash Reserves? Journal of Corporate Finance 68, Article 101924.

Chen, Zhiyao, and Ran Duchin. 2024. Do Nonfinancial Firms Use Financial Assets to Take Risk? Review of Corporate Finance Studies 13, 1–37.

Crouzet, Nicolas, and Neil Mehrotra. 2020. Small and Large Firms Over the Business Cycle. American Economic Review 110, 3549–3601.

Darmouni, Olivier, and Lira Mota. Forthcoming. The Savings of Corporate Giants. Review of Financial Studies.

Duchin, Ran, Thomas Gilbert, Jarrad Harford, and Chris Hrdlicka. 2017. Precautionary Savings with Risky Assets: When Cash Is Not Cash. Journal of Finance 72(2), 793–852.

Fabozzi, Frank, and Steven Mann. 2012. The Handbook of Fixed Income Securities. McGraw Hill; 8th edition (January 6, 2012).

Falato, Antonio, Dalida Kadyrzhanova, Jae Sim, and Roberto Steri. 2022. Rising Intangible Capital, Shrinking Debt Capacity, and the U.S. Corporate Savings Glut. Journal of Finance 77, 2799–2852.

Gao, Xiaodan, Toni M Whited, and Na Zhang. 2021. Corporate Money Demand. Review of Financial Studies 34, 1834–1866.

Gertler, Mark, and Simon Gilchrist. 1994. Monetary Policy, Business Cycles, and the Behavior of Small Manufacturing Firms. Quarterly Journal of Economics 109, 309–340.

Gross, Max, Yi Li, and Ashley Wang. 2022. Dealer Intermediation in the Primary Market of Commercial Paper. FEDS Notes. Washington: Board of Governors of the Federal Reserve

System, October 7, https://doi.org/10.17016/2380-7172.3213.

Hadlock, Charles, and Joshua Pierce. 2010. New Evidence on Measuring Financial Constraints: Moving Beyond the KZ Index. Review of Financial Studies 23(5), 1909–1940.

Huang, Teng, and Stefano Sacchetto. 2023. Bonding with Risk: Corporate Investment and Savings in Risky Financial Assets. Working paper.

Kacperczyk, Marcin, and Philipp Schnabl. 2010. When Safe Proved Risky: Commercial Paper during the Financial Crisis of 2007–2009. Journal of Economic Perspectives 24, 29–50.

Nilsen, J. H. 2002. Trade Credit and the Bank Lending Channel. Journal of Money, Credit and Banking 34(1), 226–253.

Oliner, Stephen, and Glenn Rudebusch. 1996. Monetary Policy and Credit Constraints: Evidence from the Composition of External Finance: Comment. American Economic Review 86, 300–309.

Opler, Tim, Lee Pinkowitz, Rene Stulz, and Rohan Williamson, 1999. The Determinants and´ Implications of Cash Holdings. Journal of Financial Economics 52, 3–46.

Ozdagli, Ali, and Joe Peek. 2013. Cliff Notes: The Effects of the 2013 Debt-Ceiling Crisis. Public Policy Briefs 13-9.

Stigum, Marcia, and Anthony Crescenzi. 2007. Stigum's Money Market. McGraw Hill; 4th edition (March 2, 2007).

Appendix A: Variable Definitions

Variable	Description
Cash	Cash on hand (petty cash), negotiable money orders, and demand deposits (checking accounts).
DEP	Time deposits including negotiable certificates of deposit, savings accounts, and other interest-bearing cash deposits.
CP	Commercial and finance company paper of U.S. issuers.
SLGOV	State and local government securities due in one year or less.
TRS	U.S. Treasury and federal agency securities due in one year or less or subject to agreements to sell. Include Treasury bills and securities from Federal agencies, such as notes insured by Farmers Home Administration, GNMA, or other agencies that are due in one year or less. Also include debentures and participation certificates of all Federal agencies and federally sponsored agencies including GNMA, CCC, Exim Bank, FHA, TVA, Department of Defense, Banks for Co-ops, FICB, FHLM, FHLMC, FLB, FNMA, and Postal Services with present maturities of one year or less.
Other	A residual category capturing all other short-term investments, calculated as the sum of two QFR variables, "foreign securities due in one year or less" and "other short-term financial investments." The latter variable includes bankers' acceptances, overnight deposits, and marketable securities such as stocks, commodities, and options.
STI	Total short-term investment securities defined as the sum of DEP, CP, SLGOV, TRS, and Other.
CHE	Total liquid asset holdings defined as the sum of cash and all STI securities.
Industry sigma	Standard deviation of industry cash flow divided by assets, where the standard deviation of cash flow is calculated as the standard deviation of firm-level cash flows each year averaged over each two-digit SIC code. The standard deviation of firm-level cash flows is calculated for the previous 10 years requiring at least three observations.
Log(Assets)	Log of real assets (converted to 2010 dollars).
Market to book	Market value of assets divided by book value of assets, where market value of assets is calculated as book value of assets minus book value of equity plus market value of equity.
Cash flow/Assets	Earnings after interest, dividends, and taxes but before depreciation, divided by book assets.
NWC/Assets	Net working capital net of cash, divided by book assets.
Capx/Assets	Capital expenditures divided by book assets.
Leverage	Long-term debt plus debt in current liabilities, divided by book assets.
R&D/sales	R&D expenditure divided by sales; set to zero if R&D is missing.
Dividend dummy	An indicator variable set to one if a firm pays a common dividend in a given year and zero otherwise.
Acquisition/Assets	Acquisitions divided by book assets.
Recession	An indicator variable set to one if a given quarter falls to a recessionary period according to the NBER business cycle classification and zero otherwise.
Interest Rate	Three-month Treasury security daily rate averaged each quarter.
Event2008	A dummy variable set to one if an observation falls to a period between 2008:Q3 and 2009:Q2 and zero if an observation falls to a period between 2007:Q3 and 2008:Q2.
Event2013	A dummy variable set to one if an observation falls to a period between 2013:Q4 and 2014:Q3 and zero if an observation falls to a period between 2012:Q4 and 2013:Q3.

Table 1
Summary Statistics

Panel A summarizes cash and non-cash liquid asset variables obtained from the QFR data. Panel B summarizes other firm characteristics taken from the Compustat data. The data are quarterly, spanning between 1977:Q1 and 2014:Q1. Variable definitions are provided in Appendix A.

Panel A: Cash and Short-Term Investments

Variable	N	Mean	St Dev	Median
CHE/Assets	141,000	0.112	0.150	0.047
STI/CHE	141,000	0.317	0.389	0.010
DEP/CHE	141,000	0.104	0.250	0
CP/CHE	141,000	0.059	0.189	0
TRS/CHE	141,000	0.042	0.153	0
SLGOV/CHE	141,000	0.014	0.077	0
Other/CHE	141,000	0.091	0.237	0

Panel B: Other Firm Characteristics

Variable	N	Mean	St Dev	Median
Industry sigma	99,500	0.055	0.051	0.037
Log(Assets)	99,500	6.023	1.426	5.952
Market to book	99,500	1.606	0.979	1.271
Cash flow/assets	99,500	0.018	0.051	0.021
NWC/assets	99,500	0.164	0.220	0.175
Capx/assets	99,500	0.017	0.016	0.012
Leverage	99,500	0.235	0.201	0.225
R&D/sales	99,500	0.039	0.092	0
Dividend dummy	99,500	0.524	0.499	1
Acquisition/assets	99,500	0.005	0.018	0

Table 2
Liquid Asset Compositions across Industries

This table reports average values of liquid asset variables across industries according to Fama-French 10 Industry classification (utilities and financial firms are excluded from the sample following the convention). Variable definitions are provided in Appendix A.

		Liquid Asset Component					
	N	STI/CHE	DEP/CHE	CP/CHE	TRS/CHE	SLGOV/CHE	Other/CHE
Hi-tech	24,000	0.392	0.114	0.070	0.055	0.021	0.125
Consumer durables	7,200	0.293	0.093	0.067	0.029	0.018	0.078
Consumer nondurables	17,000	0.286	0.098	0.052	0.037	0.010	0.084
Energy	9,400	0.271	0.127	0.041	0.031	0.004	0.062
Health	8,400	0.421	0.105	0.071	0.075	0.022	0.141
Manufacturing	39,000	0.328	0.120	0.061	0.038	0.014	0.087
Other	6,900	0.356	0.123	0.069	0.059	0.009	0.088
Telecom	1,400	0.313	0.097	0.049	0.087	0.006	0.074
Wholesale/retail/services	28,000	0.236	0.070	0.047	0.030	0.012	0.072

Table 3
Liquid Asset Compositions Across Firm Characteristics

This table reports average values of liquid asset variables across different firm characteristics quartiles including assets, firm age, and R&D/Sales. Variable definitions are provided in Appendix A.

Quartile	STI/CHE	DEP/CHE	CP/CHE	TRS/CHE	SLGOV/CHE	Other/CHE
Assets						
Q1 (smallest)	0.305	0.109	0.038	0.032	0.008	0.115
Q2	0.317	0.108	0.057	0.046	0.015	0.084
Q3	0.309	0.094	0.067	0.044	0.020	0.076
Q4	0.333	0.108	0.071	0.045	0.012	0.090
Q4-Q1	0.028***	-0.001	0.032***	0.013***	0.005***	-0.024***
Firm Age						
Q1 (youngest)	0.320	0.102	0.052	0.038	0.015	0.109
Q2	0.317	0.106	0.061	0.043	0.014	0.087
Q3	0.326	0.112	0.061	0.044	0.013	0.090
Q4	0.305	0.099	0.061	0.042	0.014	0.081
Q4-Q1	-0.015***	-0.003*	0.009***	0.005***	-0.001**	-0.027***
R&D/Sales						
Q1&Q2 (R&D=0)	0.270	0.092	0.050	0.037	0.010	0.075
Q3	0.333	0.123	0.067	0.036	0.012	0.088
Q4	0.398	0.116	0.069	0.056	0.022	0.126
Q4-Q1	0.128***	0.024***	0.018***	0.019***	0.012***	0.052***

Table 4
Determinants of Liquid Asset Composition

This table estimates the following specification:

$$Y_{i,t} = \alpha + \beta' X_{i,t} + \gamma_j + \theta_q + \varepsilon_{i,t},$$

where $X_{i,t}$ are the firm-level determinants of liquid asset holdings. The specification includes industry fixed effects (γ_j) and quarter fixed effects (θ_q). The dependent variable ($Y_{i,t}$) is total liquid asset holdings scaled by total assets (CHE/assets) in column (1), total short-term investment securities scaled by total liquid asset holdings (STI/CHE) in column (2), deposits relative to total liquid asset holdings (DEP/CHE) in column (3), commercial paper relative to total liquid asset holdings (CP/CHE) in column (4), state/local government securities relative to total liquid asset holdings (SLGOV/CHE) in column (5), and Treasury securities relative to total liquid asset holdings (TRS/CHE) in column (6). Standard errors are clustered at the firm level and reported in brackets. *, **, and *** represent statistical significance at the 10%, 5%, and 1% level, respectively. Variable definitions are provided in Appendix A.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Variable	CHE/assets	STI/CHE	DEP/CHE	CP/CHE	TRS/CHE	SLGOV/CHE	Other/CHE
CHE/assets		1.071***	0.139***	0.200***	0.247***	0.085***	0.366***
		[0.032]	[0.022]	[0.017]	[0.030]	[0.014]	[0.026]
Industry sigma	0.101***	-0.783***	-0.391***	-0.189***	-0.132***	-0.014	-0.034
	[0.024]	[0.076]	[0.046]	[0.032]	[0.026]	[0.013]	[0.050]
Market to book	0.022***	-0.016***	-0.009***	-0.007***	-0.004**	0.003**	0.000
	[0.002]	[0.004]	[0.003]	[0.002]	[0.002]	[0.001]	[0.003]
Log(Real Assets)	-0.015***	0.034***	0.005**	0.013***	0.010***	0.003***	0.001
	[0.001]	[0.003]	[0.002]	[0.002]	[0.001]	[0.001]	[0.002]
Cash flow/assets	0.027	0.095**	-0.023	0.02	-0.008	0.032***	0.069***
	[0.029]	[0.041]	[0.025]	[0.017]	[0.019]	[0.008]	[0.024]
NWC/assets	-0.145***	0.015	0.006	0.009	-0.005	0.001	0.006
	[0.024]	[0.020]	[0.014]	[0.007]	[0.006]	[0.003]	[0.010]
CAPX/assets	-1.288***	0.798***	-0.064	0.155*	0.036	0.080*	0.560***
	[0.080]	[0.213]	[0.168]	[0.092]	[0.084]	[0.044]	[0.120]
Leverage	-0.219***	-0.150***	-0.048***	-0.008	-0.026***	-0.027***	-0.033***
	[0.010]	[0.021]	[0.014]	[0.010]	[0.008]	[0.004]	[0.012]
R&D/Sales	0.551***	-0.199***	-0.152***	0.045*	0.008	-0.038***	-0.045
	[0.026]	[0.049]	[0.028]	[0.025]	[0.034]	[0.014]	[0.037]
Dividend dummy	-0.010***	0.052***	0.011*	0.014***	0.009**	0.004*	0.012**
	[0.003]	[0.008]	[0.006]	[0.004]	[0.004]	[0.002]	[0.006]
Acquisition/assets	-0.329***	-0.138	-0.158***	0.01	-0.023	-0.004	0.05
	[0.028]	[0.089]	[0.059]	[0.046]	[0.036]	[0.018]	[0.056]
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.470	0.195	0.029	0.041	0.075	0.061	0.061
Observations	99,500	99,500	99,500	99,500	99,500	99,500	99,500

Table 5
Within-Firm Variations

This table estimates the baseline specification with firm fixed effects and time fixed effects in place of industry fixed effects and quarter fixed effects. The dependent variable is total short-term investment securities scaled by total liquid asset holdings (STI/CHE) in column (1), deposits relative to total liquid asset holdings (DEP/CHE) in column (2), commercial paper relative to total liquid asset holdings (CP/CHE) in column (3), state/local government securities relative to total liquid asset holdings (SLGOV/CHE) in column (4), and Treasury securities relative to total liquid asset holdings (TRS/CHE) in column (5). Standard errors are clustered at the firm level and reported in brackets. *, **, and *** represent statistical significance at the 10%, 5%, and 1% level, respectively. Variable definitions are provided in Appendix A.

	(1)	(2)	(3)	(4)	(5)	(6)
Variable	STI/CHE	DEP/CHE	CP/CHE	TRS/CHE	SLGOV/CHE	Other/CHE
CHE/assets	1.034***	0.209***	0.216***	0.170***	0.068***	0.337***
	[0.037]	[0.022]	[0.021]	[0.019]	[0.010]	[0.027]
Industry sigma	0.030	0.007	-0.033	0.010	-0.018	0.077*
	[0.060]	[0.041]	[0.028]	[0.025]	[0.015]	[0.045]
Market to book	-0.012***	-0.003	-0.006***	-0.002	0.001	-0.002
	[0.004]	[0.002]	[0.002]	[0.002]	[0.001]	[0.003]
Log(Real Assets)	0.041***	0.005	0.007*	0.015***	0.006***	0.008
	[0.006]	[0.004]	[0.003]	[0.003]	[0.002]	[0.005]
Cash flow/assets	-0.028	0.003	-0.018	-0.022**	-0.007	0.016
	[0.030]	[0.020]	[0.013]	[0.011]	[0.005]	[0.017]
NWC/assets	0.001	-0.010	-0.002	-0.006	0.005	0.015
	[0.016]	[0.012]	[0.008]	[0.006]	[0.003]	[0.010]
CAPX/assets	0.377**	-0.060	0.055	-0.110	0.065*	0.383***
	[0.157]	[0.107]	[0.085]	[0.068]	[0.039]	[0.103]
Leverage	-0.168***	-0.076***	-0.021*	-0.027***	-0.014***	-0.024*
	[0.022]	[0.015]	[0.011]	[0.010]	[0.004]	[0.013]
R&D/Sales	0.002	-0.074***	0.039	0.048*	-0.013	0.008
	[0.057]	[0.027]	[0.032]	[0.028]	[0.012]	[0.041]
Dividend dummy	0.002	-0.010	-0.002	-0.003	0.003	0.012**
	[0.009]	[0.006]	[0.004]	[0.004]	[0.002]	[0.006]
Acquisition/assets	0.097	0.019	0.048	-0.022	-0.002	0.064
	[0.064]	[0.041]	[0.037]	[0.027]	[0.013]	[0.044]
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.554	0.456	0.404	0.483	0.432	0.441
Observations	99,500	99,500	99,500	99,500	99,500	99,500

Table 6
Event Regressions

This table adds event dummy variables to the baseline specification. Panel A considers the financial CP run in 2008:Q3 and compares four quarters before the run (2007:Q3–2008:Q2) and four quarters after the run (2008:Q3–2009:Q2). Panel B considers the shock to T-Bill market in 2013:Q4 and compares four quarters before (2012:Q4–2013:Q3) and the two available quarters after (data end in 2014:Q1). Other variables are not reported. Standard errors are clustered at the firm level and reported in brackets. *, **, and *** represent statistical significance at the 10%, 5%, and 1% level, respectively. Variable definitions are provided in Appendix A.

Panel A: Shock to CP Market in 2008:Q3

	(1)	(2)	(3)	(4)	(5)	(6)
Variable	STI/CHE	DEP/CHE	CP/CHE	TRS/CHE	SLGOV/CHE	Other/CHE
Event2008	-0.029***	0.012*	-0.014***	0.000	-0.006**	-0.019***
	[0.010]	[0.007]	[0.004]	[0.004]	[0.002]	[0.007]
Firm Controls	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.216	0.059	0.105	0.110	0.088	0.084
Observations	4,500	4,500	4,500	4,500	4,500	4,500

Panel B: Shock to T-Bill Market in 2013:Q4

	(1)	(2)	(3)	(4)	(5)	(6)
Variable	STI/CHE	DEP/CHE	CP/CHE	TRS/CHE	SLGOV/CHE	Other/CHE
Event2013	0.002	-0.005	0.009**	-0.007**	0.001	0.003
	[0.009]	[0.006]	[0.004]	[0.003]	[0.002]	[0.007]
Firm controls	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.213	0.075	0.109	0.153	0.099	0.096
Observations	3,900	3,900	3,900	3,900	3,900	3,900

Table 7
Macroeconomic Conditions and Short-Term Investment Compositions

This table adds macroeconomic condition measures to the baseline specification. The measure is recession dummy in panel A and interest rate in table B. Recession dummy is set to one if a given quarter falls to a recessionary period according to the NBER business cycle classification and zero otherwise. The interest rate variable is defined as three-month Treasury security daily rate averaged each quarter. Other variables are not reported. Standard errors are clustered at the firm level and reported in brackets. *, **, and *** represent statistical significance at the 10%, 5%, and 1% level, respectively. Variable definitions are provided in Appendix A.

Panel A: Adding Recession Indicator

	(1)	(2)	(3)	(4)	(5)	(6)
Variable	STI/CHE	DEP/CHE	CP/CHE	TRS/CHE	SLGOV/CHE	Other/CHE
Recession	0.002	0.019***	0.002	-0.003*	-0.004***	-0.012***
	[0.004]	[0.003]	[0.002]	[0.002]	[0.001]	[0.003]
Firm controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.54	0.448	0.4	0.478	0.427	0.437
Observations	99,500	99,500	99,500	99,500	99,500	99,500

Panel B: Adding Interest Rate

	(1)	(2)	(3)	(4)	(5)	(6)
Variable	STI/CHE	DEP/CHE	CP/CHE	TRS/CHE	SLGOV/CHE	Other/CHE
Interest rate	1.509***	0.624***	0.319***	0.275***	0.037	0.221***
	[0.112]	[0.085]	[0.059]	[0.051]	[0.026]	[0.074]
Firm controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.546	0.451	0.401	0.479	0.427	0.437
Observations	99,500	99,500	99,500	99,500	99,500	99,500

Table 8
Robustness Checks: Excluding Annual Controls

This table re-estimates the regressions in Table 5 excluding the control variables that are only available at the annual frequency. Standard errors are clustered at the firm level and reported in brackets. *, **, and *** represent statistical significance at the 10%, 5%, and 1% level, respectively. Variable definitions are provided in Appendix A.

	(1)	(2)	(3)	(4)	(5)	(6)
Variable	STI/CHE	DEP/CHE	CP/CHE	TRS/CHE	SLGOV/CHE	Other/CHE
CHE/assets	1.028***	0.207***	0.216***	0.173***	0.067***	0.332***
	[0.037]	[0.022]	[0.021]	[0.020]	[0.010]	[0.027]
Industry sigma	0.028	0.006	-0.033	0.011	-0.018	0.075*
	[0.060]	[0.041]	[0.028]	[0.025]	[0.015]	[0.045]
Market to book	-0.010***	-0.003	-0.006***	-0.003	0.002	-0.001
	[0.004]	[0.002]	[0.002]	[0.002]	[0.001]	[0.003]
Log(Real Assets)	0.042***	0.004	0.007**	0.015***	0.007***	0.009**
	[0.006]	[0.004]	[0.003]	[0.003]	[0.002]	[0.005]
Cash flow/assets	-0.026	0.014	-0.023*	-0.029**	-0.006	0.014
	[0.030]	[0.020]	[0.013]	[0.011]	[0.005]	[0.017]
NWC/assets	0.001	-0.01	-0.003	-0.006	0.005	0.016
	[0.016]	[0.012]	[0.008]	[0.006]	[0.003]	[0.010]
Leverage	-0.172***	-0.073***	-0.020*	-0.026***	-0.015***	-0.031**
	[0.022]	[0.015]	[0.012]	[0.009]	[0.004]	[0.013]
Constant	-0.014	0.075***	0.005	-0.058***	-0.032***	-0.002
	[0.040]	[0.026]	[0.022]	[0.018]	[0.010]	[0.029]
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.554	0.456	0.403	0.482	0.432	0.441
Observations	99,500	99,500	99,500	99,500	99,500	99,500

Figure 1. Liquid Assets in Compustat and the QFR

Figure 1a depicts the ratio of total liquid asset holdings to total assets between 1977:Q1 and 2014:Q1 on a quarterly frequency, averaged over all firms in the sample. The blue line and the red line show the ratio constructed using variables from Compustat and the QFR, respectively. The ratio using Compustat variables is calculated as CHEq/ATq. Similarly, Figure 1b depicts the ratio of short-term investments to total liquid asset holdings, averaged over all firms. The ratio using Compustat variables is calculated as (CHE-CH)/CHE from Annual Compustat data, as quarterly Compustat data do not provide the break-down into CH and STIs before 2007. The quarterly Compustat values are imputed as the weighted average of two nearest annual values.

Figure 1a: CHE/Total Assets

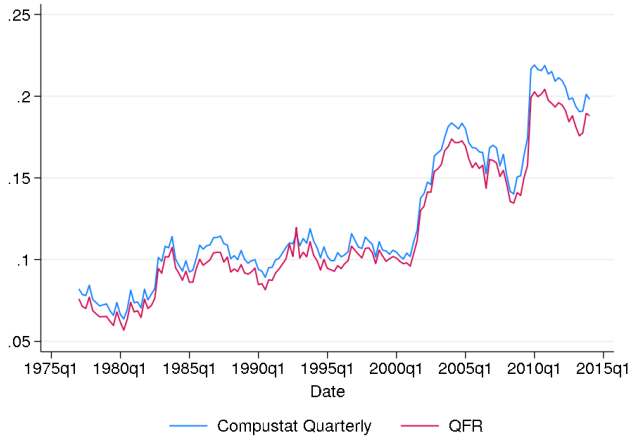


Figure 1b: STI/CHE

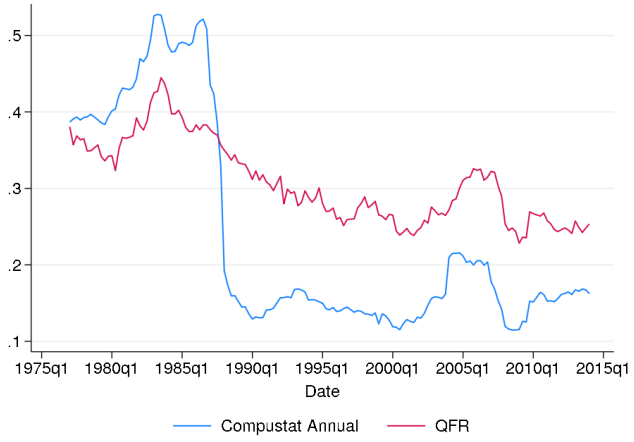


Figure 2. QFR Data in the Financial Accounts of the United States

This figure shows the nonfinancial corporate business table of the Financial Accounts of the United States, published by the Federal Reserve Board (previously known as the Flow of Funds Accounts; https://www.federalreserve.gov/Releases/Z1/DEFAULT.HTM). The variables constructed using the QFR data are highlighted in blue.

L.103 Nonfinancial Corporate Business

Billions of dollars; amounts outstanding end of period, not seasonally adjusted

			2022	2022			2023		
				Q2	Q3	Q4	Q1	Q2	
1	**FL104090005**	**Total financial assets**	**27021.6**	**26781.0**	**26072.5**	**27021.6**	**27774.8**	**28340.8**	**1**
2	FL103091003	Foreign deposits	183.4	160.6	157.7	183.4	179.4	192.5	2
3	FL103020005	Checkable deposits and currency	2036.9	1877.6	1914.2	2036.9	2000.4	2001.5	3
4	FL103030003	Time and savings deposits	285.0	264.4	285.7	285.0	263.2	367.0	4
5	FL103034000	Money market fund shares	739.5	740.1	726.5	739.5	810.1	843.6	5
6	FL102051003	Security repurchase agreements	24.7	24.6	21.2	24.7	26.2	24.8	6
7	LM104022005	Debt securities	369.0	400.0	358.1	369.0	372.6	386.6	7
8	FL103069100	Commercial paper	206.8	248.3	210.7	206.8	209.6	205.9	8
9	LM103061103	Treasury securities	90.1	81.2	79.4	90.1	90.7	104.4	9
10	LM103061703	Agency- and GSE-backed securities	30.0	27.1	26.5	30.0	30.2	34.8	10
11	LM103062003	Municipal securities	23.8	24.4	23.3	23.8	24.2	24.2	11
12	LM123063003	Corporate bonds	18.3	19.1	18.2	18.3	17.8	17.4	12
13	FL104023005	Loans	184.9	177.3	180.1	184.9	194.2	199.1	13
14	FL103065005	Mortgages	149.1	142.3	145.1	149.1	159.5	164.2	14
15	FL103066005	Consumer credit	35.8	34.9	35.0	35.8	34.7	34.9	15

Figure 3. Breakdown of Liquid Assets

The left chart breaks down total liquid asset holdings to cash (CH) and short-term investments (STIs) as a fraction of total liquid asset holdings. The right chart further breaks down STIs to five individual components, as summarized in Panel A of Table 1. Variable definitions are provided in Appendix A.

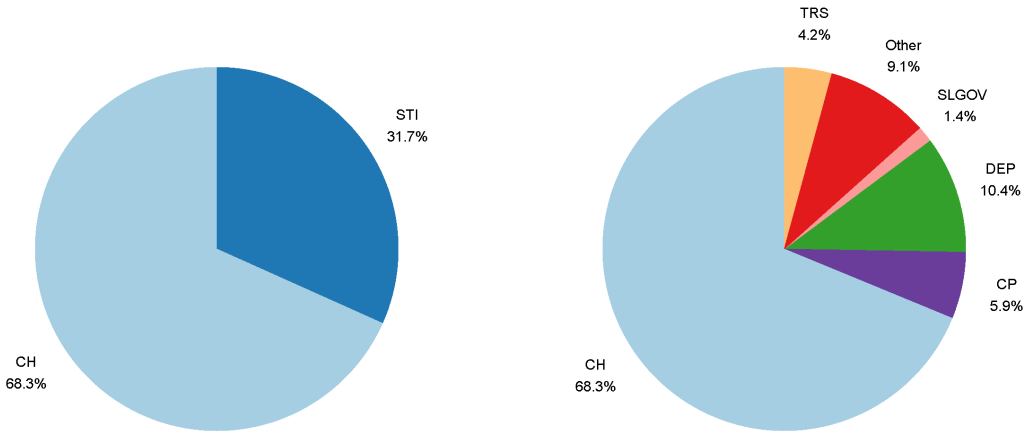


Figure 4. Composition of Short-Term Investment Securities

This figure plots the composition of short-term investment securities relative to total liquid asset holdings between 1977:Q1 and 2014:Q1 on a quarterly frequency. The area of each asset class shows the ratio of each short-term security over CHE (total liquid asset holdings), according to QFR data. Variable definitions are provided in Appendix A.

